# Statistiques avancées 3A

## CentraleSupélec

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CentraleSupélec

# Outline


1. Introduction
2. Penalized methods
3. Theoretical study of the Lasso in the linear regression model
4. Theoretical study of the Lasso in a non-linear regression model
5. Generalized Linear Model and the Lasso
6. Lasso variants in the linear regression model

# Cadre général : régression linéaire multidimensionnelle

On dispose d'un $n$-échantillon $(x_1, y_1), \ldots, (x_n, y_n)$ tel que :

$$y_i = x_i \beta^* + \varepsilon_i, \quad i = 1, \ldots, n$$

- $\varepsilon_i$ v.a. mutuellement indépendantes de loi normale $\mathcal{N}(0, \sigma^2)$, avec $\sigma$ connu
- $x_i = (x_{i,1}, \ldots, x_{i,p})$ vecteur de covariables $p$-dimensionnel, supposé déterministe
- $\beta^* \in \mathbb{R}^p$ vecteur inconnu des paramètres de régression

# Cadre général : régression linéaire multidimensionnelle

Notations :

$$X = \begin{pmatrix} x_{1,1} & \dots & x_{1,p} \\ \vdots & \ddots & \vdots \\ x_{n,1} & \dots & x_{n,p} \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} X_1 & \dots & X_p \end{pmatrix}$$

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}, \quad \beta^* = \begin{pmatrix} \beta_1^* \\ \vdots \\ \beta_p^* \end{pmatrix}$$

- $X$ : matrice des données ou matrice de design
- $X_j$ pour $j = 1, \dots, p$ : variables explicatives
- $Y$ : la réponse

# Cadre général : régression linéaire multidimensionnelle

Sous forme matricielle :

$$Y = \sum_{j=1}^{p} \beta_j^* X_j + \varepsilon = X\beta^* + \varepsilon$$

$\beta_j^*$ traduit l'influence de la variable $X_j$ sur $Y$

## La grande dimension

$$p = p(n) \geq n$$

Objectifs :

- Estimation de $\beta^*$
- Estimation de $X\beta^*$

# Les moindres carrés ordinaires

Méthode usuelle pour estimer $\beta^*$ : $\hat{\beta}$ minimise la somme des carrés résiduels

$$\sum_{i=1}^{n}(y_i - x_i\hat{\beta})^2 = \min_{\beta\in\mathbb{R}^p}\sum_{i=1}^{n}(y_i - x_i\beta)^2$$

$$\Leftrightarrow$$

$$\|Y - X\hat{\beta}\|_n^2 = \min_{\beta\in\mathbb{R}^p}\|Y - X\beta\|^2$$

## Estimateur des moindres carrés (EMC)

$$\hat{\beta}^{\mathsf{MC}} \in \operatorname*{argmin}_{\beta\in\mathbb{R}^p}\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i\beta)^2 = \operatorname*{argmin}_{\beta\in\mathbb{R}^p}\|Y - X\beta\|_n^2$$

## Théorème

Si $X^TX$ est inversible, l'EMC est unique et s'écrit

$$\hat{\beta}^{\mathsf{MC}} = (X^TX)^{-1}X^TY$$

# Erreur de prédiction de l'estimateur des moindres carrés

## Théorème

Pour tout $(p, n)$, l'erreur de prédiction de l'EMC vaut :

$$\mathbb{E}\left[\|X\hat{\beta}^{\mathsf{MC}} - X\beta^*\|_n^2\right] = \sigma^2 \frac{\mathrm{rg}(X)}{n},$$

où $\mathbb{E}$ est l'espérance sous la loi de probabilité de $\varepsilon$.

L'estimateur $X\hat{\beta}^{\mathsf{MC}}$ de $X\beta^*$, basé sur l'EMC, est consistant au sens $\ell_2$ pour l'estimation de $X\beta^*$ quand $n$ tend vers l'infini dès lors que $\frac{\mathrm{rg}(X)}{n} = o(1)$ :

- pour $p(n) = o(n)$
- $p$ fixe et $\mathrm{rg}(X) \leq \min\{p, n\}$

# Limites de l'estimateur des moindres carrés

$\beta^*$ n'est pas identifiable si la matrice $X^TX$ n'est pas inversible :

- lorsque les colonnes de $X$ sont liées
- pour $p > n$, la matrice $X^TX$ étant au plus de rang $n$ (cadre qui nous intéresse)

- EMC pas défini de manière unique et résultats obtenus non interprétables
- Estimateur en général pas consistant au sens $\ell_2$ pour l'estimation de $X\beta^*$ (sauf si les colonnes de $X$ sont liées telles que $\mathrm{rg}(X)$ asymptotiquement négligeable devant $n$)
- $p$ proche de $n$ (et $p < n$) ou variables corrélées $\Rightarrow$ valeurs propres de $X^TX$ proches de zéro $\Rightarrow$ $X^TX$ mal conditionnée et calcul numérique de l'inverse de $X^TX$ instable $\Rightarrow$ EMC inexploitable

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# Les méthodes pénalisées

$$\hat{\beta} = \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \| Y - X\beta \|_n^2 + \text{pen}(\beta, n) \right\},$$

où l'application $\text{pen} : \mathbb{R}^p \to \mathbb{R}^+$ est appelée pénalité

- le choix de la pénalité dépend de l'objectif
- bien choisie, elle permet d'éviter l'inversion de $X^T X$

# L'estimateur Ridge

## Pénalité Ridge (Hoerl et Kennard [9])

$$\text{pen}(\beta, n) = \lambda \sum_{j=1}^{p} \beta_j^2,$$

où $\lambda \geq 0$ est un paramètre de lissage qui dépend éventuellement de $n$

## Estimateur Ridge

L'estimateur Ridge $\hat{\beta}^R$ est défini de manière unique, même dans le cadre de la grande dimension, $p \geq n$, lorsque $\lambda > 0$. Il s'écrit

$$\hat{\beta}^R(\lambda) = (X^T X + \lambda I_p)^{-1} X^T Y$$

# L'estimateur Ridge

Remarque : si les $\beta_j$ ne sont pas contraints, ils peuvent prendre de très grandes valeurs et donc avoir une grande variance

Idée : pour contrôler la variance, il faut contrôler la taille des $\beta_j$

$X^TX$ et $X^TX+\kappa I_p$ ont les mêmes vecteurs propres, mais des valeurs propres différentes : si $v_j$ est un vecteur propre de $X^TX$ associé à la valeur propre $\mu_j$,

$$(X^TX+\lambda I_p)v_j = X^TXv_j+\kappa v_j = (\mu_j+\lambda)v_j$$

alors $v_j$ est un vecteur propre de $X^TX+\lambda I_p$ associé à la valeur propre $\mu_j+\lambda$

$$\hat{\beta}^R(\lambda) = (X^TX+\lambda I_p)^{-1}X^TY$$

# Propriétés de l'estimateur Ridge

- Point de vue numérique : $\hat{\beta}^R$ plus stable que $\hat{\beta}^{MC}$, choix de $\lambda$ suffisamment grand permet de rendre les valeurs propres de $X^TX + \lambda I_p$ assez grandes pour que l'inverse de cette matrice soit bien conditionnée

- Point de vue statistique : l'estimateur Ridge est biaisé

# Vers la sélection de variables

$\beta^* = (\beta_1^*, \ldots, \beta_p^*)$ traduit le poids des variables explicatives $(X_1, \ldots, X_p)$ sur la réponse $Y$

Objectif : évaluer la contribution de chaque variable et éliminer les variables non pertinentes

Plusieurs méthodes :

- Sélection de sous-ensembles (*Subset Selection*) : trouver le meilleur sous-ensemble $\mathscr{B}_k$ de taille $k$, où $k \in \{1, \ldots, p\}$ qui réduit au maximum la somme des carrés des résidus (Hastie, Tibshirani, et Friedman [8])
- Seuillage : utilisation d'un estimateur préliminaire (e.g. l'EMC si $p \leq n$), on ne sélectionne que les variables $X_j$ pour lesquelles l'estimation de $\beta_j$ dépasse un seuil arbitraire choisi par le statisticien (Donoho et Johnstone [4, 5])
- Tests : $(H_0)$ $\beta_j = 0$ contre $(H_1)$ $\beta_j \neq 0$ pour tout $j \in \{1, \ldots, p\}$ (test de student ou de Fisher si le bruit est gaussien) (Hastie, Tibshirani, et Friedman [8])

# Pénalisation $\ell_0$

$|\mathscr{A}|$ correspond au cardinal de $\mathscr{A}$, un ensemble quelconque d'indices

## Définition

Soit le modèle de régression linéaire. On définit l'ensemble de sparsité associé au vecteur $\beta^*$ par

$$\mathscr{A}^* = \{j : \beta_j^* \neq 0\}.$$

On définit l'indice de sparsité de $\beta^*$ par la quantité $|\mathscr{A}^*|$.

Objectif : construction d'estimateurs interprétables

Exemples d'estimateurs interprétables construits à partir de la pénalité $\ell_0$ :

- critère d'information du Cp de Mallows (Mallows [10])
- critère AIC (Akaike Information Criterion) (Akaike [1])
- critère BIC (Bayesian Information Criterion)(Schwartz [12])

# Pénalisation $\ell_0$

Idée générale de ces critères :

- soit $\hat{\mathscr{F}} = \{\hat{\beta}_1, \ldots, \hat{\beta}_D\}$ une collection d'estimateurs de taille $D$
- sélection dans cette collection de l'estimateur qui remplit au mieux le double objectif suivant :
  - la bonne estimation de $X\beta^*$
  - la bonne estimation des variables pertinentes $\mathscr{A}^*$
  - pénalité de ces critères :

$$\text{pen}(\beta, n) = \lambda_n \|\beta\|_0,$$

où $\|\beta\|_0 = \sum_{j=1}^{p} \mathbb{1}_{\{\beta_j \neq 0\}}$

# Cp de Mallows et AIC

Sous l'hypothèse du bruit gaussien :

$$\mathbb{E}\left[\|X\hat{\beta} - X\beta^*\|_n^2\right] + \sigma^2 = \mathbb{E}\left[\|Y - X\hat{\beta}\|_n^2\right] + \frac{2\sigma^2}{n}\mathrm{Tr}(P_{<X>})$$

où $\mathbb{E}$ désigne la covariance sous la loi de $\varepsilon$ et $P_{<X>}$ la projection sur $<X>$

$\hookrightarrow$minimiser l'erreur de prédiction équivaut à minimiser le membre de droite
$\|\beta\|_0$ est un estimateur de $Tr(P_{<X>})$

$\tilde{\beta}$ estimateur Cp et AIC est tel que

$$\|Y - X\tilde{\beta}\|_n^2 + \frac{2\sigma^2\|\tilde{\beta}\|_0}{n}$$

donne une bonne approximation de l'erreur de prédiction

$\hookrightarrow$Cp et AIC bons estimateurs pour la prédiction $X\beta^*$

# Cp de Mallows et AIC

## Modèle de régression linéaire gaussien

$$\text{pen}^{\text{AIC}}(\beta, n) = \text{pen}^{Cp}(\beta, n) := \frac{2\sigma^2 \|\beta\|_0}{n}$$

# Critère BIC

Plus approprié pour l'estimation de $\mathscr{A}^*$ car surpénalise les gros modèles

$$\text{pen}^{\text{BIC}}(\beta, n) := \sigma^2 \log(p) \frac{\|\beta\|_0}{n}$$

$\hookrightarrow$ impose une plus forte contrainte aux estimateurs dont beaucoup de composantes sont différentes de zéro, dès que $p \geq 7$

$\hookrightarrow$ tend à sélectionner des estimateurs plus parcimonieux (*sparse*) que ceux choisis par les critères Cp et AIC

# Critères Cp, AIC,BIC

## Performances théoriques :

- BIC a de meilleures performances théoriques dans le cadre de l'estimation du support $\mathscr{A}^*$ de $\beta^*$ (on parle de sélection)
- les estimateurs Cp et AIC donnent de meilleurs résultats pour l'estimation de $X\beta^*$ (on parle de prédiction).

## Inconvénient de ces critères

Complexité algorithmique trop grande dès que $D$ grand ou que $p$ grand
$\Rightarrow$ Exemple d'hypothèses sur le modèle :

- supposer les variables ordonnées
- considérer une famille $\hat{\mathscr{F}}$ d'estimateurs sparses

Ces méthodes de sélection de variables ont été introduites dans un cadre classique d'estimation, où la dimension $p$ est raisonnable ($p \leq n$)

# Hypothèse de sparsité

On se place dans un cadre de grande dimension $p > n$
Les différents objectifs :

- Estimation de $\beta^*$
- Estimation de $X\beta^*$
- Estimation du support $\mathscr{A}^*$ de $\beta^*$

## Hypothèse (S) de sparsité

Soit $\mathscr{A}^*$ l'ensemble de sparsité de $\beta^*$. Il existe un entier $s$, tel que

$$|\mathscr{A}^*| \leq s,$$

et on suppose que $s = o(n)$

$\hookrightarrow$ contrôle sur la sparsité du modèle
$\hookrightarrow$ on part du principe qu'il y a peu de variables explicatives pertinentes pour l'étude

# Estimateur des moindres carrés restreint

Cadre idéal : on connaît l'ensemble de sparsité $\mathscr{A}^*$

$\hat{\beta}^{MCR}$ estimateur des moindres carrés restreint aux variables $X_j$ dont les indices $j$ sont dans $\mathscr{A}^*$ :

$$\hat{\beta}^{MCR} = (X_{\mathscr{A}^*}^T X_{\mathscr{A}^*})^{-1} X_{\mathscr{A}^*}^T Y,$$

où $X_{\mathscr{A}^*}$ restriction de la matrice $X$ aux colonnes $X_j$ avec $j \in \mathscr{A}^*$

## Erreur de prédiction

$$\mathbb{E}\left[ \| X\hat{\beta}^{MCR} - X\beta^* \|_n^2 \right] = \sigma^2 \frac{|\mathscr{A}^*|}{n}$$

Si (S) est satisfaite, $\hat{\beta}^{MCR}$ est consistant, son erreur ne dépend pas de la dimension effective $p$ du problème, mais de la dimension réelle $|A^*|$

↪En pratique, $\mathscr{A}^*$ est rarement connu, donc $\hat{\beta}^{MCR}$ n'est pas exploitable

# Estimateur LASSO

Introduit par Tibshirani [13] :
LASSO (Least Absolute Shrinkage and Selection Operator)
Estimateur défini comme l'estimateur des moindres carrés sous une contrainte de type $\ell_1$ :

$$\tilde{\beta}^L(t) = \begin{cases} \operatorname{argmin}_{\beta \in \mathbb{R}^p} \|Y - X\beta\|_n^2, \\ \text{s.c. } \|\beta\|_1 \leq t, \end{cases}$$

où $t \geq 0$

$$\hat{\beta}^L(\lambda) \in \operatorname*{argmin}_{\beta \in \mathbb{R}^p} \left\{ \|Y - X\beta\|_n^2 + \lambda \|\beta\|_1 \right\} \tag{1}$$

Les estimateurs $\tilde{\beta}^L$ et $\hat{\beta}^L$ sont équivalents point par point :
pour un $t$ fixé, on peut trouver un $\lambda$ (qui dépend des données) tel que $\tilde{\beta}^L = \hat{\beta}^L$; pour un $\lambda$ fixé, il existe un $t$ tel que $\tilde{\beta}^L = \hat{\beta}^L$ et $t = \sum_{j=1}^p |\hat{\beta}_j^L(\lambda)|$

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# Propriétés de l'estimateur Lasso

- $\lambda = 0 \Rightarrow \hat{\beta}^L = \hat{\beta}^{MC}$ : la procédure Lasso sélectionne toutes les variables explicatives
  $\lambda \to +\infty \Rightarrow \hat{\beta}^L = \mathbf{0_p}$ : la procédure Lasso ne sélectionne aucune variable explicative
  Entre ces deux valeurs extrêmes, la procédure Lasso sélectionne d'autant plus de variables explicatives que $\lambda$ est petit, i.e., plus $\lambda$ est grand, plus la contrainte sur les coefficients de $\beta$ l'est également.
- L'estimateur Lasso peut s'interpréter en terme d'estimation dans le cadre de l'inférence bayésienne. En effet, nous pouvons déduire l'estimateur Lasso en considérant le modèle de régression avec bruit gaussien, et en prenant comme loi a priori pour le paramètre $\beta^*$ la loi de Laplace; i.e., $\beta_j^*$ admet pour densité a priori par rapport à la mesure de Lebesgue $\frac{\lambda}{2}\exp\left(-\lambda|\beta_j^*|\right)$.

# Propriétés de l'estimateur Lasso I

- L'estimateur Lasso est linéaire par morceaux comme fonction de $\lambda$.
  - Si les colonnes de $X$ sont orthogonales (ou le cas trivial où $p=1$) :

Chaque composante du vecteur $\hat{\beta}^L = (\hat{\beta}^L_1, \ldots, \hat{\beta}^L_p)^T$ s'écrit en fonction de l'estimateur des moindres carrés $\hat{\beta}^{MC} = (X^T X)^{-1} X^T Y$ :

$$\hat{\beta}^L_j(\lambda) = \mathrm{sgn}(\hat{\beta}^{MC}_j)(|\hat{\beta}^{MC}_j| - \lambda/2)_+, \quad \forall j \in \{1, \ldots, p\}$$

où $(\alpha)_+ = \max(\alpha, 0)$ pour $\alpha$ un réel et la fonction $\mathrm{sgn}(.)$ est définie pour tout réel $b$ par

$$\mathrm{sgn}(b) = \begin{cases} 1 & \text{si } b > 0 \\ 0 & \text{si } b = 0 \\ -1 & \text{si } b < 0 \end{cases}$$

Si on choisit $\lambda^{(k)} = |\hat{\beta}^{MC}_k|$, $k \in \{1, \ldots, p\}$, la linéarité de $\hat{\beta}^L$ change en fonction de ces $\lambda$

# Propriétés de l'estimateur Lasso II

- Si $X$ n'est pas une matrice orthogonale (et $p \neq 1$) :
  les valeurs du paramètre de régularisation telles que la linéarité change peuvent être trouvées en considérant les conditions d'optimalité du premier ordre (1), les conditions Karush-Kuhn-Tucker (KKT).
  Ensemble de sparsité ou ensemble actif de l'estimateur Lasso $\hat{\beta}^L(\lambda)$:

$$\hat{\mathscr{A}}_L = \hat{\mathscr{A}}(\lambda) = \{j \in \{1, \dots, p\} : \hat{\beta}_j^L \neq 0\}$$

  Conditions KKT :

$$\begin{cases} 2(X^j)^T(Y - X\hat{\beta}^L(\lambda)) = \lambda \operatorname{sgn}(\hat{\beta}_j^L) & \forall j \in \hat{\mathscr{A}}_L, \\ 2|(X^j)^T(Y - X\hat{\beta}^L(\lambda))| \leq \lambda & \forall j \notin \hat{\mathscr{A}}_L \end{cases} \tag{2}$$

  Pour trouver des valeurs du paramètre de régularisation $\{\lambda^{(1)}, \dots, \lambda^{(K)}\}$ pour lesquels la linéarité de $\hat{\beta}^L(\lambda)$ change, on considère les conditions non saturées dans les conditions KKT ($2^{\text{ème}}$ ligne de (2)) : pour tout $j \notin \hat{\mathscr{A}}_L$, il existe $\lambda^{(k)}$ tel que $|(X^j)^T(Y - X\hat{\beta}^L(\lambda^{(k)}))| = \lambda^{(k)}/2$.
  Pour $\lambda = \lambda^{(k)}$, la contrainte correspondant à $X^j$ devient saturée et $X^j$ devient active (i.e., l'indice $j$ "passe" de $\hat{\mathscr{A}}_L^c$ à $\hat{\mathscr{A}}_L$).

# Propriétés de l'estimateur Lasso III

- Une solution approchée de l'estimateur Lasso peut être obtenue grâce par exemple à l'algorithme LARS (Efron, Hastie, Johnstone, et Tibshirani [6])
  Algorithme basée sur les conditions KKT, i.e., sur les corrélations entre le résidu $Y - X\hat{\beta}^L(\lambda)$ et les variables explicatives $(X_1, \ldots, X_p)$.

  Inconvénient de cet algorithme :
  → échoue à construire de bonnes évaluations de l'estimateur lorsque de fortes corrélations interviennent entre les variables
  → tend à sélectionner une variable explicative par groupe de variables corrélées, écartant ainsi de l'étude les autres variables du groupe

# Nature des résultats

3 objectifs différents :

- fournir la meilleure approximation du vecteur $X\beta^*$ : on parle de prédiction
- estimer le vecteur $\beta^*$ : on parle d'estimation
- identifier le support $\mathscr{A}^*$ de $\beta^*$, ou encore le vecteur signe de $\beta^*$ : on parle de sélection

# Nature des résultats

## Définition

Soit l'hypothèse $(S)$ de sparsité vérifiée et $s$ l'indice associé. On dit qu'un estimateur $\hat{\beta}$ vérifie une Inégalité de Sparsité (IS)

en prédiction si : $$\mathbb{E}\left[\|X\beta^* - X\hat{\beta}\|_n^2\right] \leq Cs\frac{L(n,p)}{n}$$

en estimation en norme $\ell_1$ si : $$\mathbb{E}\left[\|\hat{\beta} - \beta^*\|_1\right] \leq Cs\sqrt{\frac{L(n,p)}{n}}$$

en estimation en norme-sup si : $$\mathbb{E}\left[\|\hat{\beta} - \beta^*\|_\infty\right] \leq C\sqrt{\frac{L(n,p)}{n}}$$

où $C > 0$ et $L(n,p)$ dépendance au plus logarithmique en $n$ et $p$.

Remarque : Toutes les IS peuvent être également définies à l'aide d'inégalités probabilistes.

# Nature des résultats

IS généralement difficiles à obtenir $\Rightarrow$ nécessité d'ajouter une hypothèse, souvent restrictive, sur la matrice de Gram, $\Psi^n = \dfrac{X^T X}{n}$

# Nature des résultats

IS généralement difficiles à obtenir $\Rightarrow$ nécessité d'ajouter une hypothèse, souvent restrictive, sur la matrice de Gram, $\Psi^n = \dfrac{X^T X}{n}$

"Bon" estimateur :

- Optimalité : résultats d'optimalité de deux types :
  - d'un point de vue asymptotique ($n \to +\infty$), consistance de la procédure d'estimation.
  - d'un point de vue non-asymptotique, on recherche des IS du même type que celles décrites dans la Définition précédente
- Hypothèses : hypothèses sur la matrice de Gram $\Psi^n$ aussi faibles que possible.
- Complexité algorithmique : méthode d'estimation utilisable en pratique; i.e., implémentable en un temps raisonnable même lorsque $p$ est plus grand que $n$
- Interprétabilité : estimateur sparse $\Rightarrow$ possibilité d'extraire l'information utile et d'interpréter les résultats

# Résultats théoriques pour l'estimateur Lasso[1]

[1] Bickel, Peter J and Ritov, Ya'acov and Tsybakov, Alexandre B, *Simultaneous analysis of Lasso and Dantzig selector*, The Annals of Statistics [2]

# Notations

- $\mathscr{M}(\beta) = \sum_{j=1}^{p} \mathbb{1}_{\{\beta_j \neq 0\}} = |J(\beta)|$ number of nonzero coordinates of $\beta$
  $\rightarrow$ characterizes the sparsity of the vector $\beta$: the smaller $\mathscr{M}(\beta)$, the "sparser" $\beta$
- $J(\beta) = \{j \in \{1, \ldots, p\} : \beta_j \neq 0\}$ and $|J|$ the cardinality of $J$
- For a vector $\delta \in \mathbb{R}^p$ and a subset $J \subset \{1, \ldots, p\}$, $\delta_J$ is the vector in $\mathbb{R}^p$ that has the same coordinates as $\delta$ on $J$ and zero coordinates on the complement $J^c$ of $J$

## Lasso estimator

$$\hat{\beta}^L = \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \frac{1}{n} \| Y - X\beta \|_2^2 + 2r \|\beta\|_1 \right\}, \quad r > 0 \tag{3}$$

# Assumption on the Gram matrix I

Assumption on the Gram matrix $\Psi^n$ allows only a weak correlation between variables

Notations:

- Matrice de Gram:

$$\Psi^n = \left[\Psi^n_{j,k}\right]_{1\leq j,k\leq p} = \frac{X^T X}{n},$$

- $\phi_{max}$ maximal eigenvalue of $\Psi^n$

**Assumption**

The diagonal elements of the Gram matrix $\Psi^n$ are all equal to 1.

**Remark:** this assumption is just for the sake of simplicity.

# Assumption on the Gram matrix II

## Degenerate case :

For $p > n$ and even $p \gg n$, $\Psi^n$ is degenerate

$$\Rightarrow \min_{\delta \in \mathbb{R}^p : \delta \neq 0} \frac{(\delta^T \Psi^n \delta)^{1/2}}{\|\delta\|_2} = \min_{\delta \in \mathbb{R}^p : \delta \neq 0} \frac{\|X\delta\|_2}{\sqrt{n}\|\delta\|_2} = 0 \tag{4}$$

$\Rightarrow$ ordinary least squares does not work in this case, since it requires positive definiteness of $\Psi^n$

$\rightarrow$ Lasso requires much weaker assumptions: the minimum in (4) can be replaced by the minimum over a restricted set of vectors, and the norm $\|\delta\|_2$ in the denominator of the condition can be replaced by the $\ell_2$–norm of only a part of $\delta$

# Assumption on the Gram matrix III

The residuals of the linear regression model $\delta = \hat{\beta}^L - \beta^*$ satisfy with probability close to 1:

$$\|\delta_{J_0^c}\|_1 \leq 3\|\delta_{J_0}\|_1, \qquad \text{where } J_0 = J(\beta^*) \tag{5}$$

Let $\Psi$ be a positive definite $(p \times p)-$matrix, $\omega_n = \max_{i,j} |(\Psi^n - \Psi)_{i,j}|$ the maximal difference between the elements of $\Psi^n$ and $\Psi$; and $\delta$ satisfies (5)

$$\begin{aligned}\frac{\delta^T \Psi^n \delta}{\|\delta\|_2^2} &= \frac{\delta^T \Psi \delta + \delta^T (\Psi^n - \Psi)\delta}{\|\delta\|_2^2} \geq \frac{\delta^T \Psi \delta}{\|\delta\|_2^2} - \frac{\omega_n \|\delta\|_1^2}{\|\delta\|_2^2} \\ &\geq \frac{\delta^T \Psi \delta}{\|\delta\|_2^2} - \omega_n \left(\frac{4\|\delta_{J_0}\|_1}{\|\delta_{J_0}\|_2}\right)^2 \geq \frac{\delta^T \Psi \delta}{\|\delta\|_2^2} - 16\omega_n |J_0|\end{aligned}$$

$\Rightarrow$ for $\omega_n |J_0|$ small enough, $\dfrac{\delta^T \Psi^n \delta}{\|\delta\|_2^2}$ bounded away from 0

$\Rightarrow$ kind of "restricted" positive definiteness, which is valid only for the vectors satisfying (5).

# Assumption on the Gram matrix IV

## Restricted Eigenvalue (RE) assumptions : $\mathrm{RE}(s,3)$

For some integer $s$ such that $1 \leq s \leq p$, the following condition holds:

$$\kappa(s,3) = \min_{\substack{J_0 \subseteq \{1,\dots,p\}, \\ |J_0| \leq s}} \min_{\substack{\delta \neq 0, \\ \|\delta_{J_0^c}\|_1 \leq 3\|\delta_{J_0}\|_1}} \frac{\|X\delta\|_2}{\sqrt{n}\|\delta_{J_0}\|_2} > 0$$

Remark:

- $s$ plays the role of an upper bound on the sparsity $\mathscr{M}(\beta^*)$
- under Assumption $\mathrm{RE}(s,3)$,

$$\min\{\|X\delta\|_2 : \mathscr{M}(\delta) \leq 2s, \delta \neq 0\} > 0$$

$\Rightarrow$ the square submatrices of size $\leq 2s$ of the Gram matrix are necessarily positive definite

# Theoretical results

## Theorem (Bickel, Ritov, Tsybakov (2009))

Assumptions :

- $\varepsilon_i$ independent $\mathcal{N}(0,\sigma^2)$ random variables with $\sigma^2 > 0$,
- the diagonal elements of the matrix $X^T X/n$ equal to 1,
- $\mathcal{M}(\beta^*) \leq s$, where $1 \leq s \leq p$, $n \geq 1$, $p \geq 2$,
- $\mathrm{RE}(s,3)$ is satisfied.

Consider the Lasso estimator $\hat{\beta}^L$ defined by (3) with $r = A\sigma\sqrt{\frac{\log p}{n}}$ and $A > 2\sqrt{2}$. Then, with probability at least $1 - p^{1-A^2/8}$, we have

$$\|\hat{\beta}^L - \beta^*\|_1 \leq \frac{16A}{\kappa^2(s,3)}\sigma s\sqrt{\frac{\log p}{n}}, \tag{6}$$

$$\|X(\hat{\beta}^L - \beta^*)\|_2^2 \leq \frac{16A^2}{\kappa^2(s,3)}\sigma^2 s\log p, \tag{7}$$

$$\mathcal{M}(\hat{\beta}^L) \leq \frac{64\phi_{max}}{\kappa^2(s,3)}s. \tag{8}$$

# Theoretical results

**Remarks:**

- Gives only non-asymptotic upper bounds on the loss, with some probability and under some conditions. The probability depends on $p$ and the conditions depend on $n$ and $p$. Recall that Assumption $\mathrm{RE}(s,3)$ is imposed on the $n \times p$ matrix $X$. To deduce asymptotic convergence (as $n \to +\infty$ and/or as $p \to +\infty$), we would need some very strong additional properties.
- Assumption $\mathrm{RE}(s,3)$ does not implies identifiability of $\beta^*$, but $\beta^*$ is uniquely defined because of the assumption $\mathscr{M}(\beta^*) \leq s$

# Outline


1. Introduction
2. Penalized methods
3. Theoretical study of the Lasso in the linear regression model
4. Theoretical study of the Lasso in a non-linear regression model
5. Generalized Linear Model and the Lasso
6. Lasso variants in the linear regression model

# Non-linear regression model

The model :

$$Y_i = f(Z_i) + \varepsilon_i, \quad i = 1, \ldots, n$$

- $Z_i \in \mathbb{R}^p$ deterministic covariates
- $\varepsilon_1, \ldots, \varepsilon_n$ i.i.d. $\mathcal{N}(0, \sigma^2)$
- $f : \mathbb{R}^p \to \mathbb{R}$ regression function to be estimated

# Estimation in high-dimensional non-linear regression models

$\mathscr{F} = \{f_1, \ldots, f_M\}$ set of functions called a dictionary,
$M$ can be much larger than $n$

Objective: learn from data an estimator of $f$ of the form

$$f_\beta(z) = \sum_{j=1}^{M} \beta_j f_j(z)$$

with the following properties:

- the empirical error $\frac{1}{n}\sum_{i=1}^{n}(f_\beta(Z_i) - f(Z_i))^2$ is small
- the sparsity of $\beta$: $\|\beta\|_0 = |J(\beta)| = |\{j = 1, \ldots, M : \beta_j \neq 0\}|$ is small compared to $M$

If both properties are satisfied $\Rightarrow$ we can find a sparse linear combination of elements of $\mathscr{F}$ to estimate $f$

$\Rightarrow$ Sparse estimation

# Estimation in high-dimensional non-linear regression models

Remark: if $M = p$ and $f_j(x) = x_j$, we recover the standard linear regression model where $f_\beta(x) = x^T\beta$

Notations:

$$X = \begin{pmatrix} f_1(Z_1) & \dots & f_M(Z_1) \\ \vdots & & \vdots \\ f_1(Z_n) & \dots & f_M(Z_n) \end{pmatrix} = \begin{pmatrix} X_1^T \\ \vdots \\ X_n^T \end{pmatrix} = \begin{pmatrix} X^1 & \dots & X^M \end{pmatrix}$$

$$y = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad f = \begin{pmatrix} f(Z_1) \\ \vdots \\ f(Z_n) \end{pmatrix}, \quad f_\beta = \begin{pmatrix} f_\beta(Z_1) \\ \vdots \\ f_\beta(Z_n) \end{pmatrix}, \quad \varepsilon = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

Gaussian linear model:

$$y = X\beta + \varepsilon,$$

$M \gg n \Rightarrow X$ can be overdetermined: it is not full-rank

$\|u\|$ Euclidean norm of $u \in \mathbb{R}^n$

# Estimation in high-dimensional non-linear regression models

Oracle inequalities: we are looking for an estimator $\hat{\beta}_n$ s.t. $\|\hat{\beta}_n\|_0 \ll M$ and

$$\frac{1}{n}\|f_{\hat{\beta}_n} - f\|^2 \leq \inf_{\beta \in \mathbb{R}^M} \left\{ \frac{1}{n}\|f_\beta - f\|^2 + remainder(\beta) \right\} \quad \text{oracle inequality}$$

ideally, $remainder(\beta)$ small quantity that depends on $\beta$, but might depends on $n, \mathscr{F}, \sigma^2$

Interpretation:

If *remainder* is small, then such an inequality would prove that the estimator $f_{\hat{\beta}_n}$ performs almost as well as the best linear combination $f^* = f_{\beta^*}$ of elements from $\mathscr{F}$, where $\beta^* \in \underset{\beta \in \mathbb{R}^M}{\arg\min} \|f_\beta - f\|^2$

$f^*$ is an oracle since it depends on the unknown function $f$

# Estimation in high-dimensional non-linear regression models

Questions:

- How can we construct a sparse estimator $\hat{\beta}_n$?
- What is the value of *remainder* in the oracle inequality?

$$\hat{\beta}_n^{lasso} = \underset{\beta \in \Theta}{\arg\min} \left\{ \frac{1}{n} \sum_{i=1}^{n} (Y_i - f_\beta(X_i))^2 + \lambda \|\beta\|_1 \right\}, \quad \Theta \subset \mathbb{R}^M \text{ convex set}$$

# Convex optimization

Convex relaxation: $\ell_1$ convex relaxation of $\ell_0$

Convex envelope[2] of $g_0(x) = \mathbb{1}_{\{x \neq 0\}}$ over the interval $[-1,1]$ is $g_1(x) = |x|$

$\Rightarrow$ Convex envelope of $x \mapsto \|x\|_0$ over $[-1,1]^M$ is $x \mapsto \|x\|_1$

Constrained optimization problem:

$$\begin{cases} \min \|x\|_0 \\ \text{s.t. } x \in C \text{ and } \|x\|_\infty \leq R, \text{ where } C \text{ is a convex set} \end{cases}$$

$$\Leftrightarrow \begin{cases} \min \mathbb{1}^T u \\ \text{s.t. } u \in \{0,1\}^M, |x_i| \leq R u_i \text{ for all } i = 1, \ldots, M \\ x \in C \end{cases}$$

$\hookrightarrow$ hard to solve exactly since it requires to try out all possible vectors $u \in \{0,1\}^M$

[2] convex envelope of $g : [a,b] \to \mathbb{R}$ is at each point $x \in [a,b]$ the supremum of all convex functions that lie under $g$: $g^{env}(x) = \sup\{h(x) : h \text{ convex and } h \leq g \text{ over} [a,b]\}$

# Convex optimization

## Convex relaxation of this problem:

$$\begin{cases} \min \mathbb{1}^T u \\ \text{s.t. } u \in \{0,1\}^M, |x_i| \leq R u_i \text{ for all } i = 1, \dots, M \\ x \in C \end{cases}$$

$$\Leftrightarrow \begin{cases} \min \frac{1}{R} \|x\|_1 \\ \text{s.t. } x \in C \text{ and } \|x\|_\infty \leq R \end{cases}$$

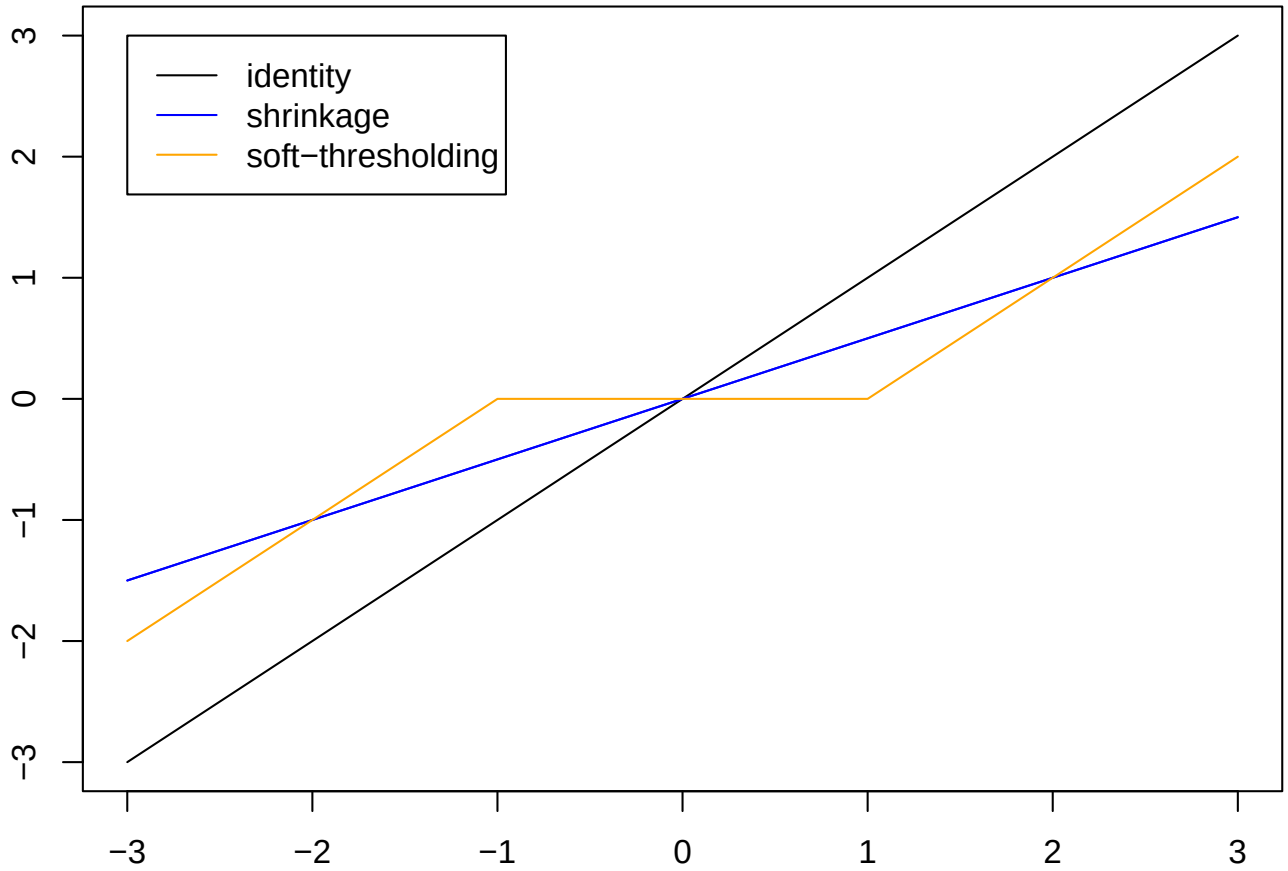
## Soft-thresholding:

$$\operatorname*{argmin}_{a \in \mathbb{R}} \left\{ \frac{1}{2}(a-b)^2 + \lambda |a| \right\} = \operatorname{sgn}(b)(|b| - \lambda)_+, \quad \text{for any } b \in \mathbb{R}$$

$$\Rightarrow \operatorname*{argmin}_{a \in \mathbb{R}^M} \left\{ \frac{1}{2}\|a-b\|_2^2 + \lambda \|a\|_1 \right\} = T_\lambda(b), \quad \forall b \in \mathbb{R}^M,$$

where $(T_\lambda(b))_j = \operatorname{sgn}(b_j)\left(|b_j| - \lambda\right)_+$, for $j = 1, \dots, M$

# Convex optimization

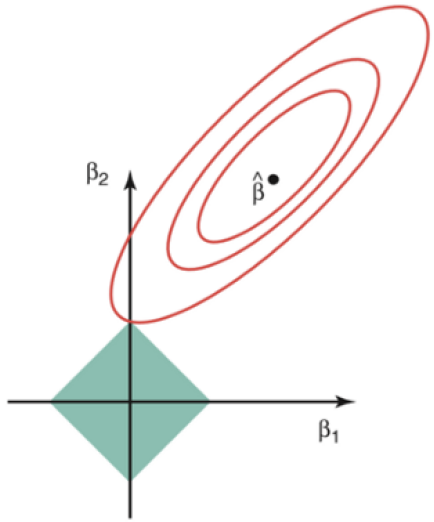



Soft-thresholding and shrinkage with $\lambda = 1$ on a single coordinate

# Convex optimization

Interpretation:

- shrinkage operator (= ridge regression) does not induce sparsity while soft-thresholding does
- $\ell_1-$norm induces sparsity because of the fact that the absolute value is not differentiable at 0
- geometrically, unit $\ell_1$ ball has sparse corners at $\pm e_j$ for $j = 1, \ldots, M$ (canonical basis vectors) that are sparse vectors: when we project a point onto an $\ell_1-$ball, we are likely to project onto a corner or an edge, that are sets of sparse points



# Some tools from convex optimization

$\phi : \mathbb{R}^p \to \mathbb{R}$ convex function

## Definition: Subdifferential

$g \in \mathbb{R}^p$ is a subgradient of a convex function $\phi : \mathbb{R}^p \to \mathbb{R}$ at $u \in \mathbb{R}^p$ if and only if

$$\phi(v) - \phi(u) \geq g^T (v - u) \quad \text{for any } v \in \mathbb{R}^p$$

The subdifferential of $\phi$ at $u$ is defined by

$$\partial\phi(u) = \left\{ g \in \mathbb{R}^p : \phi(v) - \phi(u) \geq g^T (v - u) \text{ for all } v \in \mathbb{R}^p \right\}$$

$\hookrightarrow$ it generalizes the differential to non-differentiable convex functions

Example: $\phi(u) = |u|$,

$$\partial\phi(u) = \{1\} \text{ if } u > 0$$
$$\partial\phi(u) = \{-1\} \text{ if } u < 0$$
$$\partial\phi(u) = [-1, 1] \text{ if } u = 0$$

# Some tools from convex optimization

## Properties

- $u^* \in \underset{u \in \mathbb{R}^p}{\arg\min} \phi(u)$ if and only if $0 \in \partial\phi(u^*)$
- $\partial\left(\sum_{k=1}^{K} \alpha_k \phi_k(u)\right) = \sum_{k=1}^{K} \alpha_k \partial\phi_k(u)$ whenever $\alpha_k \geq 0$ and $\phi_k$ convex functions for all $k = 1, \ldots, K$
- if $\phi(u) = \max_{k=1,\ldots,K} \phi_k(u)$, $\partial\phi(u) = conv\left(\bigcup_{k=1}^{K} \{\partial\phi_k(u) : \phi_k(u) = \phi(u)\}\right)$

  $conv(A)$ : convex hull of a set $A$

# Some tools from convex optimization

$C \subset \mathbb{R}^p$ convex set, $\delta_C(u) = \begin{cases} 0 & \text{if } u \in C \\ +\infty & \text{if } u \notin C \end{cases}$

$$u^* \in \underset{u \in C}{\arg\min}\ \phi(u) \Leftrightarrow u^* \in \underset{u \in \mathscr{R}^p}{\arg\min} \left\{\phi(u) + \delta_C(u)\right\}$$
$$\Leftrightarrow 0 \in \partial\phi(u^*) + \partial\delta_C(u^*)$$
$$\Leftrightarrow \exists g^* \in \partial\phi(u^*) : -g^* \in \partial\delta_C(u^*)$$

$g \in \partial\delta_C(u^*)$ with $u^* \in C \Leftrightarrow \delta_C(u) - \underbrace{\delta_C(u^*)}_{=0} \geq g^T(u - u^*),\ \forall u \in \mathbb{R}^p$

$\Rightarrow \forall u \in C, \quad \partial\delta_C(u) = \left\{g \in \mathbb{R}^p : g^T(v - u) \leq 0, \forall v \in C\right\}$ normal cone to $C$ at $u$

# Some tools from convex optimization

## Proposition

Let $\phi : \mathbb{R}^p \to \mathbb{R}$ be a convex function and $C \subset \mathbb{R}^p$ a convex set. An optimally criterion for the problem $u^* \in \underset{u \in C}{\arg\min}\ \phi(u)$ is given by

$$\exists g^* \in \partial\phi(u^*) \text{ s.t. } (g^*)^T(u - u^*) \geq 0, \text{for all } u \in C$$

## Proposition: monotonicity of the subdifferential

Given a convex function $\phi : \mathbb{R}^p \to \mathbb{R}$, any $u_1, u_2 \in \mathbb{R}^p$ and any $g_1 \in \partial\phi(u_1)$ and $g_2 \in \partial\phi(u_2)$, we have

$$(u_1 - u_2)^T(g_1 - g_2) \geq 0$$

# Some tools from convex optimization

## Subdifferential of $\ell_1$–norm:

$$\phi(u) = \|u\|_1 = \max\left\{e^T u : e \in \{-1,1\}^p\right\} = \max_{i=1,\dots,2^p} \phi_i(u),$$

where $\phi_i(u) = e_i^T u$ and $e_1, \dots, e_{2^p} \in \{-1,1\}^p$

We define $I(u) = \left\{i \in \{1,\dots,2^p\} : e_i^T u = \|u\|_1\right\}$. Each $\phi_i$ is differentiable and $\nabla \phi_i(u) = e_i$

$$\begin{aligned}\partial \|u\|_1 &= conv\left(\bigcup_{i \in I(u)} \{e_i\}\right) \\ &= \left\{\mathrm{sgn}(u) + h, h \in \mathbb{R}^p, \|h\|_\infty \leq 1, h \odot u = 0\right\},\end{aligned}$$

where $h \odot u$ is the Hadamard product: $(h \odot u)_j = h_j u_j$

Example: if $p = 4$, $u = \begin{pmatrix}17 & -42 & 0 & 3\end{pmatrix}^T$, then

$$\partial \|u\|_1 = \{1\} \times \{-1\} \times [-1,1] \times \{1\}$$

# Slow oracle inequality I

## Assumption

The columns of $X$ are standardized $\|X^j\|_2 = \sqrt{n}$.

**Remark:** unrestrictive assumption that follows good practice when using linear methods in ML

$$\hat{\beta}_n = \underset{\beta \in \Theta}{\arg\min} \left\{ \frac{1}{n} \sum_{i=1}^{n} (Y_i - f_\beta(Z_i))^2 + \lambda \|\beta\|_1 \right\}$$

$$= \underset{\beta \in \Theta}{\arg\min} \left\{ \frac{1}{n} \|y - f_\beta\|^2 + \lambda \|\beta\|_1 \right\} \quad \text{for some convex set } \Theta \subset \mathbb{R}^p,$$

where $\lambda = 2\sigma \sqrt{\frac{2(x + \log M)}{n}}$,
$x > 0$ confidence level and $\sigma > 0$ standard deviation of the noise

# Slow oracle inequality II

**Theorem**

If $\hat{\beta}_n$ is the Lasso estimator, we have that

$$\frac{1}{n}\|f_{\hat{\beta}_n} - f\|^2 \leq \inf_{\beta \in \Theta} \left\{ \frac{1}{n}\|f_\beta - f\|^2 + 2\lambda\|\beta\|_1 \right\}$$

with probability larger than $1 - 2e^{-x}$.

$\hookrightarrow$ slow oracle inequality: the remainder is $O(1/\sqrt{n})$

Proof

# Fast oracle inequality I

In order to obtain a faster $O(1/n)$ rate, we need an extra assumption on the matrix of features $X$:

$$G = \frac{1}{n} X^T X = \left[ \frac{1}{n} \langle f_j, f_{j'} \rangle \right]_{1 \leq j, j' \leq M}, \quad f_j = [f_j(Z_1), \ldots, f_j(Z_n)] \in \mathbb{R}^n$$

If $M > n$, we have that

$$\min_{t \in \mathbb{R}^M \setminus \{0\}} \frac{\sqrt{t^T G t}}{\| t \|} = \min_{t \in \mathbb{R}^M \setminus \{0\}} \frac{\| Xt \|}{\sqrt{n} \| t \|} = 0$$

since $X : \mathbb{R}^M \to \mathbb{R}^n$ and $Ker(X) \neq \{0\}$, hence the smallest eigenvalue of $G$ is $0$

# Fast oracle inequality II

## Assumption: Restricted Eigenvalues

$X$ satisfies the $\mathrm{RE}(s, c_0)$ assumption for some $c_0 > 0$ and some $s \in \{1, \dots, M\}$ whenever:

$$\kappa(s, c_0) = \min_{\substack{J \subset \{1,\dots,M\} \\ |J| \leq s}} \min_{\substack{\delta \neq 0 \\ \|\delta_{J^c}\|_1 \leq c_0 \|\delta_J\|_1}} \frac{\|X\delta\|_2}{\sqrt{n}\|\delta_J\|_2} > 0$$

## Proposition

If $X$ satisfies $\mathrm{RE}(s, 1)$, then any submatrix of $X$ formed by any subset of $2s$ columns from $X$ has full rank.

# Fast oracle inequality III

## Theorem

Under $\mathrm{RE}(s,3)$, we have

$$\frac{1}{n}\|f_{\hat{\beta}_n} - f\|^2 \leq \inf_{\substack{\beta\in\Theta \\ \|\beta\|_0 \leq s}} \left\{ \frac{1}{n}\|f_\beta - f\|^2 + \frac{18\sigma^2}{\kappa(s,3)^2}\frac{s(x+\log M)}{n} \right\}$$

with a probability larger than $1-2\mathrm{e}^{-x}$.

$\hookrightarrow$ the Lasso estimator realizes a balance between an approximation term $\|f_\beta - f\|^2$ and a complexity term which involves the sparsity of $\beta$

$\hookrightarrow$ the Lasso estimator, which is solution of a simple convex problem, is almost as good as the best sparse representation of $f$ using the dictionary $\mathscr{F}$

$\hookrightarrow$ in this inequality, the convergence rate is of order $\frac{s\log M}{n}$, where $s$ is the sparsity of the best $\beta$

# Outline


1. Introduction
2. Penalized methods
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# Introduction

Generalized Linear Models (GLM) [Nelder and Wedderburn (1972)]:

- include gaussian linear, logit and log-linear models
- express the expectation of the variable of interest $Y$ as a function of a linear combination of the covariates

# The exponential family

## Definition

We say that the random variable $Y$ belongs to exponential family if its density can be written as follows:

$$f_\theta(y) = \exp\left(\frac{y\theta - b(\theta)}{a(\phi)} + c(y,\phi)\right),$$

where $a, b, c$ are deterministic functions such that $c$ is differentiable, $b$ three times differentiable and its first derivative is invertible. $\theta$ is the natural parameter and $\phi$ is the dispersion parameter, both unknown.

Examples:

1) Normal distribution $\mathcal{N}(\mu, 1)$:

$$f_\mu(y) = \text{ à compléter}$$

2) Bernoulli distribution $\mathcal{B}(p)$:

$$f_p(y) = \text{ à compléter}$$

# Definition of the GLM

A Generalized Linear Model for the data $(x_1, Y_1), \ldots, (x_n, Y_n)$ consists in

1) an exponential model for $Y \in \mathbb{R}$
2) a linear predictor $\eta_i = x_i^T \beta$, (with $\beta \in \mathbb{R}^p$)
3) a link function such that $g(\mathbb{E}(Y_i)) = x_i^T \beta$

**Remark:**

- This model allows to link $Y$ with the linear predictor $\eta_i = x_i^T \beta$, assuming that $\mu_i = \mathbb{E}(Y_i) = g^{-1}(\eta_i) = g^{-1}(x_i^T \beta)$ for some link function $g$.
- The link between the linear predictor and the exponential model is through the expectation of $Y$ and the link function $g$.
- There are many possible choices for the function $g$, but it must take all the values that $\mathbb{E}(Y)$ can handle. However, there is a particular choice for the link function $g$: the canonical link function.

The link function $g$ is assumed to be monotonic and differentiable.

# The link function

## Definition

Let $Y$ be a random variable with a density function $f_\theta(y) = \exp(Q(\theta)y - A(\theta))h(y)$, then the canonical link function is the function $g$ such that $\theta = g(\mu)$, i.e., $\theta = g(E(Y))$.

**Remark:** When we choose the canonical link function, we have $\theta = g(\mu) = x^T\beta$, i.e., $\theta_i = g(\mu_i) = x_i^T\beta$.

# Examples I

$Y_1, \ldots, Y_n$ $n$ independent variables with distribution $f_{\theta_i}$ belonging to the exponential family, where $\theta_i$ is the parameter of the exponential family

1) Normal distribution $\mathcal{N}(\mu_i, \sigma^2)$:

$$f_{\mu_i}(y_i) = \exp\left\{-\frac{1}{2}\frac{\mu_i^2}{\sigma^2}\right\} \exp\left\{-\frac{1}{2}\frac{y_i^2}{\sigma^2} - \frac{1}{2}\ln(2\pi\sigma^2)\right\} \exp\left\{y_i \frac{\mu_i}{\sigma^2}\right\}$$

$$\theta_i = \mathbb{E}(Y_i) = \mu_i \quad \text{and} \quad g(x) = x$$

2) Bernoulli distribution $\mathcal{B}(\pi_i)$:

$$f_{\pi_i}(y_i) = \pi_i^{y_i}(1-\pi_i)^{1-y_i} = (1-\pi_i)\exp\left\{y_i \ln\frac{\pi_i}{1-\pi_i}\right\}$$

$$\theta_i = \ln\frac{\pi_i}{1-\pi_i} \quad \text{and} \quad g(x) = \ln\left(\frac{x}{1-x}\right)$$

This relationship defines the logit function for the canonical link function associated with this model: $logit(x) = \ln\left(\frac{x}{1-x}\right)$

# Examples II

3) Poisson distribution $\mathscr{P}(\lambda_i)$:

$$f_{\lambda_i}(y_i) = \frac{\lambda_i^{y_i} e^{-\lambda_i}}{y_i!} = \exp(-\lambda_i)\frac{1}{y_i!}\exp(y_i \ln \lambda_i)$$

$$\theta_i = \ln \lambda_i \quad \text{and} \quad g(x) = \ln(x)$$

# Moments

$\ell(\theta_i, \phi; y_i) = \ln\ f_{\theta_i}(y_i; \phi)$: the contribution of the i$^{\text{th}}$ observation to the log-likelihood

$$\ell(\theta_i, \phi; y_i) = \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)$$

We have

$$\frac{\partial \ell}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{a(\phi)} \quad \text{and} \quad \frac{\partial^2 \ell}{\partial \theta_i^2} = -\frac{b''(\theta_i)}{a(\phi)}$$

Under regularity assumptions (differentiation under the integral),

$$\mathbb{E}\left(\frac{\partial \ell}{\partial \theta}\right) = 0 \quad \text{and} \quad -\mathbb{E}\left(\frac{\partial^2 \ell}{\partial \theta^2}\right) = \mathbb{E}\left[\left(\frac{\partial \ell}{\partial \theta}\right)^2\right].$$

Then,

$$\mathbb{E}[Y_i] = b'(\theta_i) \quad \text{and} \quad \text{Var}(Y_i) = b''(\theta_i)a(\phi)$$

# Estimation I

Recall that: $\eta_i = x_i^T \beta = g(\mu_i)$ where $\mu_i = \mathbb{E}(Y_i)$
The estimators of the $\beta_j$ are calculated by maximizing the log-likelihood.
Log-likelihood:

$$\mathscr{L}(\beta) = \sum_{i=1}^{n} \ln f_{\theta_i}(y_i; \phi) = \sum_{i=1}^{n} \ell(\theta_i, \phi; y_i)$$

We have:

$$\frac{\partial \ell_i}{\partial \beta_j} = \frac{\partial \ell_i}{\partial \theta_i} \frac{\partial \theta_i}{\partial \mu_i} \frac{\partial \mu_i}{\partial \eta_i} \frac{\partial \eta_i}{\partial \beta_j}$$

# Estimation II

$$\frac{\partial \ell_i}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{a(\phi)} = \frac{y_i - \mu_i}{a(\phi)}$$

$$\frac{\partial \mu_i}{\partial \theta_i} = b''(\theta_i) = \frac{\mathrm{Var}(Y_i)}{a(\phi)}$$

$$\frac{\partial \eta_i}{\partial \beta_j} = x_{ij}$$

$\frac{\partial \mu_i}{\partial \eta_i}$ depend on the link function $\eta_i = g(\mu_i)$

Likelihood equations:

$$\sum_{i=1}^{n} \frac{(y_i - \mu_i) x_{ij}}{\mathrm{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = 0, \quad j = 1, \ldots, p$$

$\rightsquigarrow$ Non linear equations in $\beta$ that need iterative methods to be solved

# Particular case: the canonical link function

In that case : $\eta_i = \theta_i = x_i^T \beta$ and

$$\frac{\partial \mu_i}{\partial \eta_i} = \frac{\partial \mu_i}{\partial \theta_i} = \frac{\partial b'(\theta_i)}{\partial \theta_i} = b''(\theta_i)$$

Then,

$$\frac{\partial \ell_i}{\partial \beta_j} = \frac{y_i - \mu_i}{\mathrm{Var}(Y_i)} b''(\theta_i) x_{ij} = \frac{y_i - \mu_i}{a(\phi)} x_{ij}$$

If $a(\phi)$ is constant for the observations, the equations become: $X^T y = X^T \mu$
In the gaussian case with $\mu = X\beta$ (the link function is the identity), we obtain the solution

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

which corresponds to the ordinary least squares estimator

# Resolution of the likelihood equations

## Iterated Reweighted Least Squares (IRLS)

To calculate (approximate) the Maximum Likelihood Estimator, we use a Newton-Raphson type algorithm.

At iteration $k$, we denote $\hat{\beta}^k$ the current solution. We approximate $\ell_n = \frac{1}{n}\ell$ by a quadratic function:

$$\ell_n(\hat{\beta}^k + h) = \ell_n(\hat{\beta}^k) + \nabla \ell_n(\hat{\beta}^k)^T h + \frac{1}{2} h^T \nabla^2 \ell_n(\hat{\beta}^k) h$$

we minimize this approximation to obtain $h^*$, we consider

$$\hat{\beta}^{k+1} = \hat{\beta}^k + h^*$$

and we iterate.

# Theoretical results

We introduce $\mathscr{I}(\gamma) = -\mathbb{E}\left(\nabla^2 \ell_n(\gamma)\right)$ for all $\gamma$.

## Consistency and asymptotic normality

Under certain conditions (cf. Fahrmeier and Kaufmann [7]), we can prove that for every true parameter $\beta$,

- $|\hat{\beta} - \beta| \xrightarrow{\mathbb{P}} 0$
- $\hat{\beta}$ is asymptotically gaussian

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{\mathscr{L}} \mathcal{N}(0, \mathscr{I}(\beta)^{-1})$$

- and also

$$\sqrt{n}(\mathscr{I}(\hat{\beta}))^{1/2}(\hat{\beta} - \beta) \xrightarrow{\mathscr{L}} \mathcal{N}(0, I_p)$$

This results are satisfied for the considered models (with canonical link function), when the covariates are bounded and if $\lambda_{\min}(X^T X) \to +\infty$.

Proof

# Outline


1. Introduction
2. Penalized methods
3. Theoretical study of the Lasso in the linear regression model
4. Theoretical study of the Lasso in a non-linear regression model
5. Generalized Linear Model and the Lasso
6. Lasso variants in the linear regression model

# The bridge estimator

- For $\lambda \geq 0$ and $\gamma \geq 0$, we set

$$C_{\lambda,\gamma}(\beta) := \|Y - X\beta\|^2 + \lambda \|\beta\|_\gamma^\gamma$$

with $\|\beta\|_\gamma^\gamma = \begin{cases} \sum_{j=1}^p |\beta_j|^\gamma, & \text{if } \gamma > 0 \\ \sum_{j=1}^p \mathbb{1}_{\{\beta_j \neq 0\}} & \text{if } \gamma = 0 \end{cases}$ and

$$\hat{\beta}_{\text{Bridge}}(\lambda, \gamma) = \underset{\beta \in \mathbb{R}^p}{\arg\min}\ C_{\lambda,\gamma}(\beta),$$

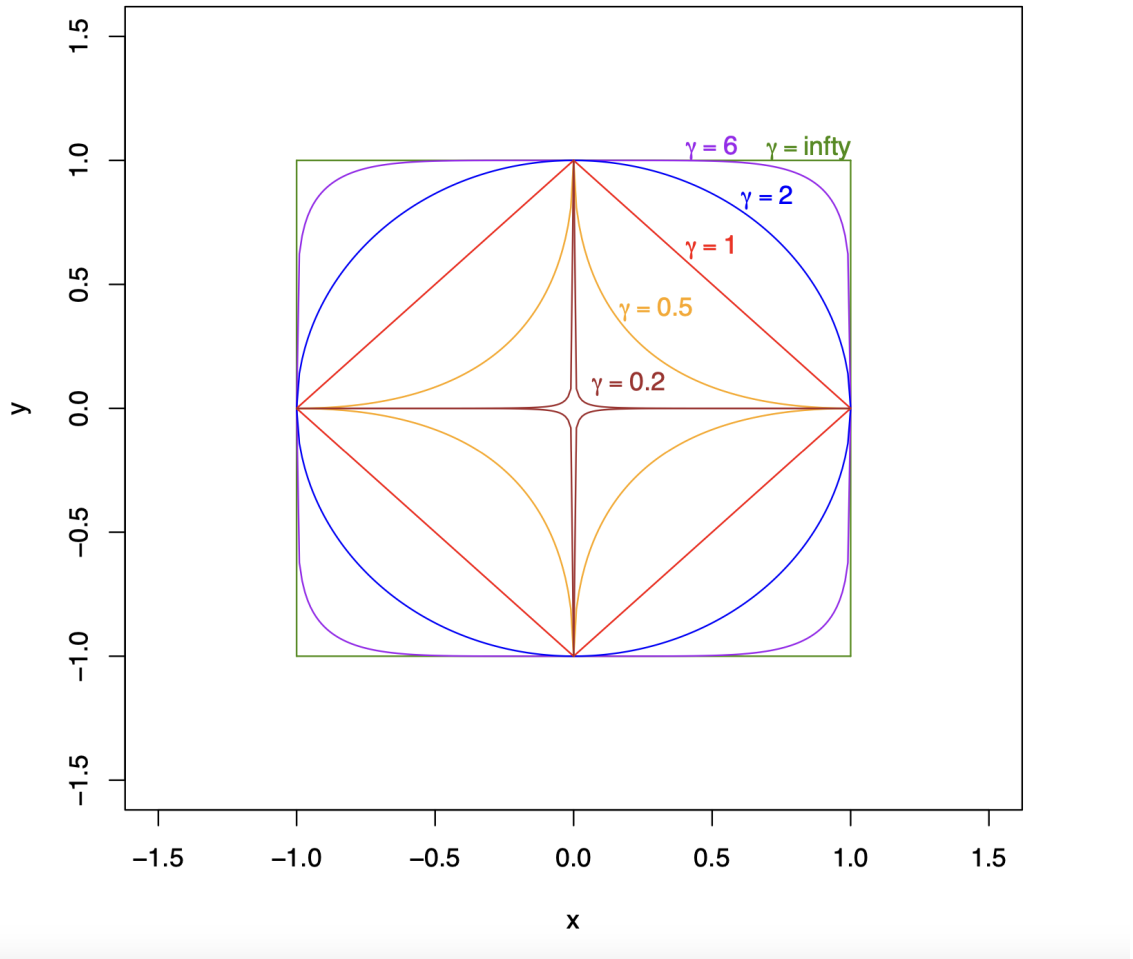
- Three interesting cases $(\lambda > 0)$:
  1. $\gamma = 0$: model selection
  2. $\gamma = 2$: Ridge estimation
  3. $\gamma = 1$: Lasso estimation

# The bridge estimator

- Assume that $\gamma = 0$. Then the bridge estimate exists if $X$ is one-to-one.
- Assume that $\gamma > 0$. Then the bridge estimate exists.
- If $0 \leq \gamma < 1$,then $C_{\lambda,\gamma}$ is not convex and it may be very hard to minimize it in high dimensions.
- Assume that $\gamma = 1$. The penalized criterion $C_{\lambda,1}$ is then convex and $C_{\lambda,1}$ has one minimizer if $X$ is one-to-one.
- Assume that $\gamma > 1$. The penalized criterion $C_{\lambda,\gamma}$ is then strictly convex and $C_{\lambda,\gamma}$ has only one minimizer. Almost surely, all coordinates of the bridge estimate are non-zero.
- Equivalence with the constrained problem

$$\hat{\beta}_{\text{Bridge}}(\lambda,\gamma) = \underset{\{\beta \in \mathbb{R}^p : \|\beta\|_\gamma^\gamma \leq t\}}{\arg\min} \|Y - X\beta\|^2.$$

# The bridge estimator[3]




Constraints regions $\|\beta\|_\gamma^\gamma$ for different values of $\gamma$. The region is convex if and only if $\gamma \geq 1$.

[3]Vincent Rivoirard, Statistics for high-dimensional data

# Graphical illustration for $p = 2$

- We take $X^T X = \begin{pmatrix} 4 & 1.4 \\ 1.4 & 1 \end{pmatrix}$ and $t = 1$.
- Note that

$$\| Y - X\beta \|^2 = (\beta - \hat{\beta}^{\text{ols}})^T X^T X (\beta - \hat{\beta}^{\text{ols}}) + \| Y - X\hat{\beta}^{\text{ols}} \|^2$$

and the constrained problem becomes

$$\underset{\{\beta \in \mathbb{R}^p : \|\beta\|_\gamma^\gamma \leq t\}}{\arg\min} \left\{ (\beta - \hat{\beta}^{\text{ols}})^T X^T X (\beta - \hat{\beta}^{\text{ols}}) \right\}.$$

- We compare the Ridge estimate

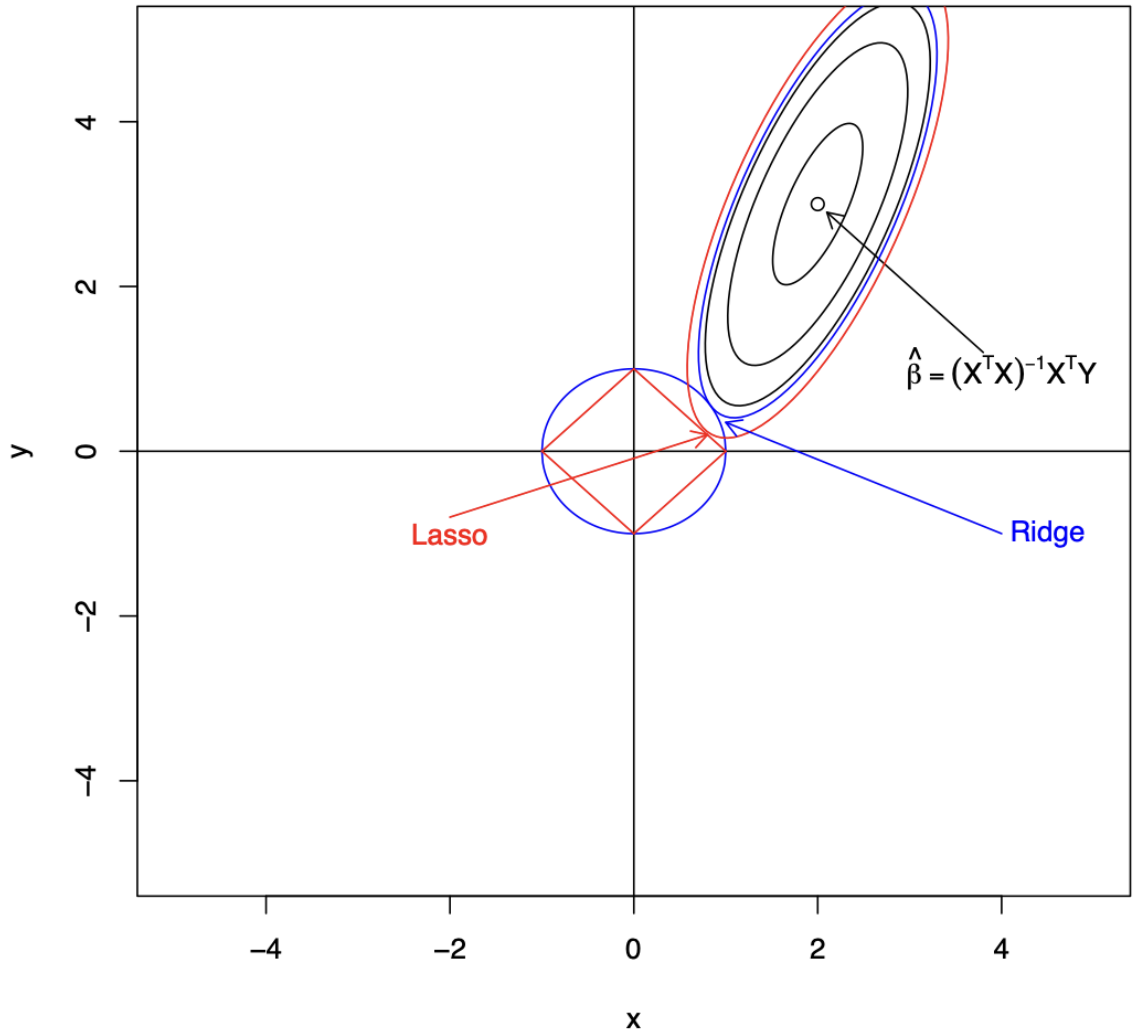
$$\hat{\beta}^{\text{ridge}}(\lambda) := \underset{\{\beta \in \mathbb{R}^p : \|\beta\|_2^2 \leq t\}}{\arg\min} \left\{ \{ (\beta - \hat{\beta}^{\text{ols}})^T X^T X (\beta - \hat{\beta}^{\text{ols}}) \right\}$$

and the Lasso estimate

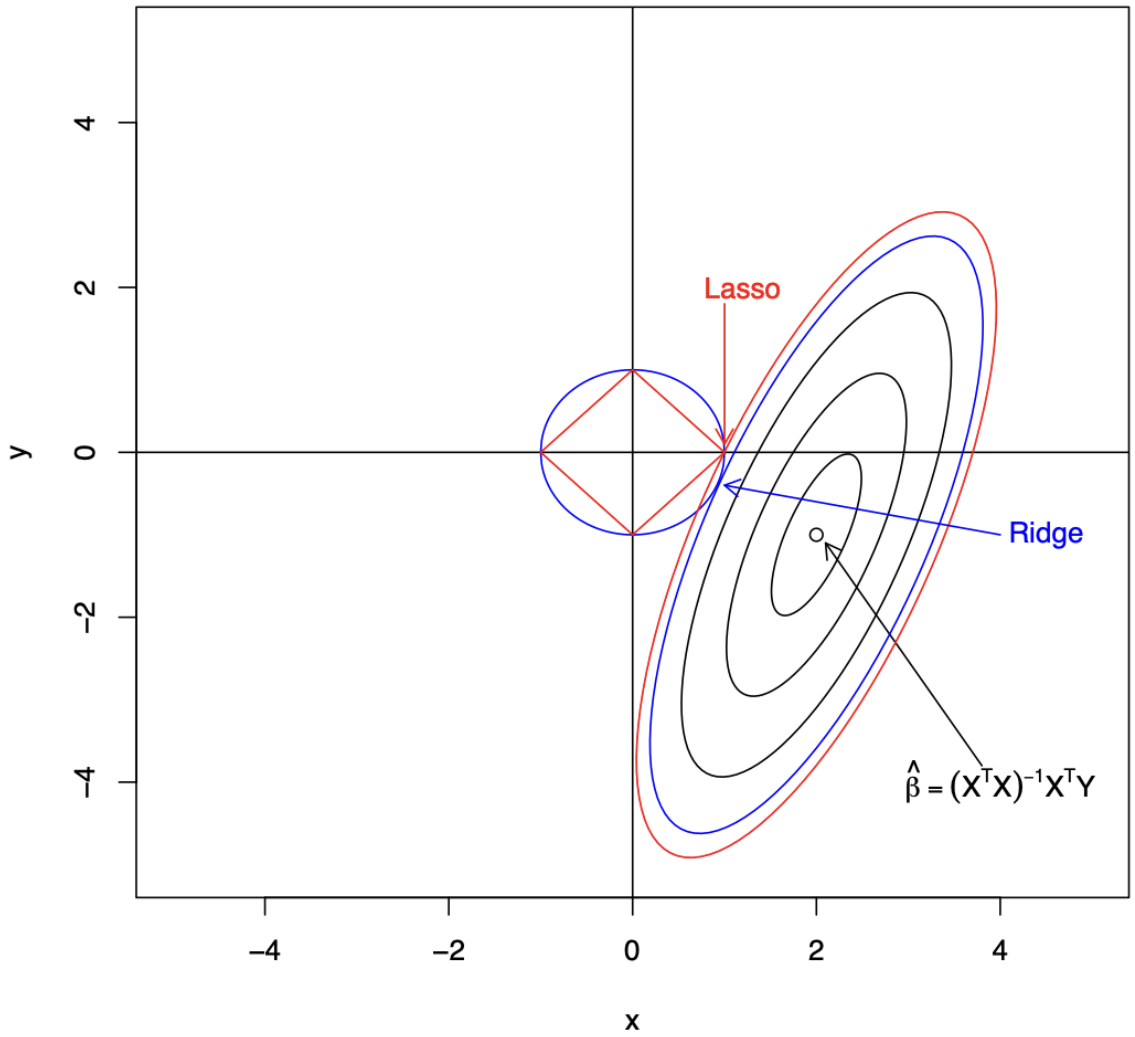
$$\hat{\beta}^{\text{ridge}}(\lambda) := \underset{\{\beta \in \mathbb{R}^p : \|\beta\|_1 \leq t\}}{\arg\min} \left\{ \{ (\beta - \hat{\beta}^{\text{ols}})^T X^T X (\beta - \hat{\beta}^{\text{ols}}) \right\}$$

- Of course both estimates are close (for same values of $t$) but, depending on $\hat{\beta}^{\text{ols}}$, Lasso estimate may have null coordinates.

# Graphical illustration for $p = 2$



# Graphical illustration for $p = 2$



# A two step procedure: Lasso+Least-squares (LSLasso[4])

- Problem: the Lasso is biased (large coefficients are shrunk to zero)
- A two-step procedure

**Algorithm 1** Least Squares Lasso (LSLasso)

1. Lasso procedure to obtain : $\hat{\beta}^L(\lambda)$
2. Least squares on the active covariates ($\in \text{supp}(\hat{\beta}^L(\lambda))$

$$\hat{\beta}^{\text{LSLasso}}(\lambda) = \underset{\beta \in \mathbb{R}^p:\ \text{supp}(\beta) = \text{supp}(\hat{\beta}^L(\lambda))}{\arg\min} \frac{1}{2} \| Y - X\beta \|_2^2$$

⚠ Cross Validation must be performed on the entire procedure

**Remark:** implemented in the **glmnet** package (argument *relax*= *T*)

[4] Lederer, J. (2013). Trust, but verify: benefits and pitfalls of least-squares refitting in high dimensions. arXiv preprint arXiv:1306.0113.

# The Dantzig selector

- Remember that the Lasso estimate satisfies the constraint

$$\|2X^T(Y - X\hat{\beta}_L(\lambda)\|_\infty \leq \lambda$$

  We then introduce the convex set

$$\mathscr{D}_\lambda = \{\beta \in \mathbb{R}^p : \|2X^T(Y - X\beta)\|_\infty \leq \lambda\}$$

  called the Dantzig constraint
- Remember also that we investigate sparse vectors where sparsity is measured using the $\ell_1-$norm
- Therefore, Candès and Tao (2007) [3] have suggested to use the Dantzig selector

$$\hat{\beta}_D(\lambda) := \underset{\beta \in \mathscr{D}_\lambda}{\arg\min} \ \|\beta\|_1$$

- Note that $\|\hat{\beta}_D(\lambda)\|_1 \leq \|\hat{\beta}_L(\lambda)\|_1$. Numerical and theoretical performances of Dantzig and Lasso estimates are very close. In some cases, they may even coincide.

# Adaptive Lasso

- Due its "soft-thresholding nature", the Lasso estimation of large coefficients may suffer from a large bias. We can overcome this problem by introducing data-driven weights.
- Zou (2006) [16] proposed an adaptive version of the classical Lasso

$$\hat{\beta}_{\text{AdapL}}(\lambda) := \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \| Y - X\beta \|^2 + \lambda \sum_{j=1}^{p} \omega_j |\beta_j| \right\}$$

with

$$\omega_j = \frac{1}{|\tilde{\beta}_j|},$$

and $\tilde{\beta}$ is a preliminary estimator. Usual choice if $p$ is not too large : $\omega_j = \frac{1}{|\hat{\beta}_j^{ols}|}$, with $\hat{\beta}^{ols}$ the ordinary least squares estimator

- The larger $|\tilde{\beta}_j|$, the smaller $\omega_j$, which encourages large values for $(\hat{\beta}_{\text{AdapL}}(\lambda))_j$

# Relaxed Lasso

- Instead of introducing weights, Meinshausen (2007) [11] suggests a two-step procedure:
  1. Compute
  $$\hat{\beta}_L(\lambda) = \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \|Y - X\beta\|^2 + \lambda \|\beta\|_1 \right\}$$
  and set
  $$\hat{S}_\lambda := \left\{ j \in \{1, \dots, p\} : \hat{\beta}_{L,j}(\lambda) \neq 0 \right\}$$
  2. For $\delta \in [0, 1]$,
  $$\hat{\beta}_{\text{relaxedL}}(\lambda, \delta) := \underset{\beta \in \mathbb{R}^p,\ J(\beta) \subset \hat{S}_\lambda}{\arg\min} \left\{ \|Y - X\beta\|^2 + \delta\lambda \|\beta\|_1 \right\}$$
- If $X$ is orthogonal,
$$\hat{\beta}_{\text{relaxedL},j}(\lambda, \delta) = \begin{cases} X_j^T Y - \dfrac{\delta\lambda}{2} & \text{if } X_j^T Y \geq \dfrac{\lambda}{2} \\ 0 & \text{if } -\dfrac{\lambda}{2} \leq X_j^T Y \leq \dfrac{\lambda}{2} \\ X_j^T Y + \dfrac{\delta\lambda}{2} & \text{if } X_j^T Y \leq -\dfrac{\lambda}{2} \end{cases}$$
- The value $\delta = 0$ is commonly used.

# Elastic net I

Two drawbacks of the Lasso :

- In the p>n case, the lasso selects at most n variables before it saturates
- In the model $Y = X\beta + \varepsilon$, consider

$$\hat{\beta}_L(\lambda) = \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \| Y - X\beta \|^2 + \lambda \| \beta \|_1 \right\}$$

If we consider $\tilde{X} = [X, X_p] \in \mathbb{R}^{n \times p}$ and if $\hat{\beta}_{L,p}(\lambda) \neq 0$, then any vector $\tilde{\beta}(\lambda)$ such that

$$\tilde{\beta}_j(\lambda) := \begin{cases} \hat{\beta}_{L,j}(\lambda) & \text{if } j \in \{1, \ldots, p-1\} \\ \alpha \hat{\beta}_{L,p}(\lambda) & \text{if } j = p \\ (1-\alpha)\hat{\beta}_{L,p}(\lambda) & \text{if } j = p+1 \end{cases}$$

with $\alpha \in [0,1]$ is solution of $\underset{\beta \in \mathbb{R}^{p+1}}{\arg\min} \left\{ \| Y - \tilde{X}\beta \|^2 + \lambda \| \beta \|_1 \right\}$

$\hookrightarrow$ infinite number of solutions.

# Elastic net II

If there is a group of variables among which the pairwise correlations are very high, then the Lasso tends to select only one variable from the group and does not care which one is selected.

- In practice, predictors are different but they may be strongly correlated. In this case, the Lasso estimate may hide the relevance of one of them, just because it is highly correlated to another one. Coefficients of two correlated predictors should be close.
- The elastic net procedure proposed by Zou and Hastie (2005) [17] makes a compromise between Ridge and Lasso penalties: given $\lambda_1 > 0$ and $\lambda_2 > 0$,

$$\hat{\beta}_{e.n}(\lambda_1, \lambda_2) := \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \|Y - X\beta\|^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|^2 \right\}$$

The criterion is strictly convex, so there is a unique minimizer.

- We can improve $\hat{\beta}_{e.n}(\lambda_1, \lambda_2)$ and consider $(1+\lambda_2)\hat{\beta}_{e.n}(\lambda_1, \lambda_2)$ (see Zou and Hastie (2005) [17])

# Fused Lasso

- For change point detection, for instance, for which coefficients remain constant over large portions of segments, Tibshirani, Saunders, Rosset, Zhu and Knight (2005) [14] have introduced the fused Lasso: given $\lambda_1 > 0$ and $\lambda 2 > 0$,

$$\hat{\beta}_{\text{fused}}(\lambda_1, \lambda_2) := \underset{\beta \in \mathbb{R}^p}{\operatorname{argmin}} \left\{ \|Y - X\beta\|^2 + \lambda_1 \sum_{j=1}^{p} |\beta_j| + \lambda_2 \sum_{j=2}^{p} |\beta_j - \beta_{j-1}| \right\}$$

- The first penalty is the familiar Lasso penalty which regularizes the signal. The second penalty encourages neighboring coefficients to be identical.
- We can generalize the notion of neighbors from a linear ordering to more general neighborhoods, for examples adjacent pixels in image. This leads to a penalty of the form

$$\lambda_2 \sum_{j \sim j'} |\beta_j - \beta_{j'}|.$$

- Parameters $\lambda_1$ and $\lambda_2$ are hard to tune.

# The group-Lasso

- To select simultaneously a group of variables, Yuan and Lin (2006) [15] suggest to use the group-Lasso procedure : we are given $K$ known non-overlapping and non-empty groups $G_1, G_2, \ldots, G_K$, for $\lambda > 0$,

$$\hat{\beta}^{\text{group}}(\lambda) := \underset{\beta \in \mathbb{R}^p}{\arg\min} \left\{ \|Y - X\beta\|^2 + \lambda \sum_{k=1}^{K} \|\beta_{(k)}\| \right\}$$

  where $\beta_{(k)_j} = \beta_j$ if $j \in G_k$ and 0 otherwise.
- If $K = p$, $\|\beta_{(k)}\| = |\beta_k|$ and the Group-Lasso is the standard Lasso.
- As for the Lasso, the group-Lasso can be characterized: For any $k \in \{1, \ldots, K\}$,

$$\begin{cases} 2X_{(k)}^T (Y - X\hat{\beta}^{\text{group}}) = \lambda \dfrac{\hat{\beta}_{(k)}^{\text{group}}}{\|\hat{\beta}_{(k)}^{\text{group}}\|_2} & \text{if } \hat{\beta}_{(k)}^{\text{group}} \neq 0 \\ \left\| 2X_{(k)}^T (Y - X\hat{\beta}^{\text{group}}) \right\| \leq \lambda & \text{if } \hat{\beta}_{(k)}^{\text{group}} = 0 \end{cases}$$

- The procedure keeps or discards all the coefficients within a block and can increase estimation accuracy by using information about coefficients of the same block.

# Conclusion I

- To address the computational complexity associated with model selection, one can consider using convex criteria, particularly leading to Lasso-type estimates.
- This induces some bias, but a reduction of the variance in predicted values. Moreover, it allows for the identification of a small set of predictors with the most significant effects, which simplifies interpretation.
- By varying the basic Lasso $\ell_1-$penalty, we can reduce problems encountered by the standard Lasso or incorporate some prior knowledge about the model.
- In the context of linear regression, these easily computable estimates are highly popular for high-dimensional statistics and exhibit favorable theoretical and numerical properties.

# Conclusion II

- Despite standard techniques for tuning the Lasso, its calibration remains a significant ongoing challenge.
- Lasso type estimates have been detailed in the linear regression setting and apply in the Generalized Linear Models. But it can be extended to other settings: for latent variable models, density estimation, counting processes, etc.

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